

Experiment 3

Aim: To analyse the performance of a multi class confusion matrix by using choosen database with python code

Program

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from sklearn.datasets import load_digits
from sklearn.model_selection import train_test_split
from sklearn.ensemble import RandomForestClassifier
from sklearn.metrics import confusion_matrix
import seaborn as sns
import matplotlib.pyplot as plt
from sklearn.metrics import accuracy_score, precision_score, recall_score, f1_score
X, y= load_digits(return_X_y=True)
X_train, X_test, y_train, y_test = train_test_split(X, y, test_size=0.25)
clf = RandomForestClassifier(random_state=23)
clf.fit(X_train, y_train)
y_pred = clf.predict(X_test)
cm = confusion_matrix(y_test, y_pred)
sns.heatmap(cm,
annot=True,
fmt='g', cmap="winter")
plt.ylabel('Prediction', fontsize=13)
plt.xlabel('Actual', fontsize=13)
plt.title('Confusion Matrix', fontsize=17)
plt.show()
accuracy = accuracy_score(y_test, y_pred)
print("Accuracy :", accuracy)
```

Output:

